Course Title: Cryptography and Network Security Lab	**Course Code**: BCSE309P
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EX1A

CLASSICAL CRYPTOGRAPHIC ALGORITHM

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Code: (--CyptoFunctions--)

Cypto_functions.py:

```
def caesar_encrypt(text, key):
    encrypted_text = ""
    for char in text:
        if char.isupper():
            encrypted_text += chr( (ord(char) - ord('A') + key) %26 + ord('A'))
        elif char.islower():
            encrypted_text += chr( (ord(char) - ord('a') + key) %26 + ord('a'))
        else:
            encrypted_text+=char
    return encrypted_text

def caesar_decrypt(text, key):
    decrypt_text = ""
    for char in text:
        if char.isupper():
            decrypt_text += chr( (ord(char) - ord('A') -key) %26 + ord('A'))
        elif char.islower():
            decrypt_text += chr( (ord(char) - ord('a') -key) %26 + ord('a'))
        else:
            decrypt_text+=char
    return decrypt_text

def vigenere_encrypt(text, key):
    encrypted = ""
    key = key.lower()
    key_index = 0

    for char in text:
        if char.isalpha():
            shift = ord(key[key_index % len(key)]) - ord('a')

            if char.isupper():
                encrypted += chr( (ord(char) + shift - ord('A')) % 26 + ord('A'))
            else:
                encrypted += chr( (ord(char) + shift - ord('a')) % 26 + ord('a'))
            key_index+=1
        else:
            encrypted += char
    return encrypted

def vigenere_decrypt(text, key):
    decrypt = ""
    key = key.lower()
    key_index = 0
```

```python
    for char in text:
        if char.isalpha():
            shift = ord( key[key_index % len(key)] ) - ord('a')
            if char.isupper():
                decrypt+= chr( (ord(char) - ord('A') - shift) % 26 + ord('A'))
            else:
                decrypt+= chr( (ord(char) - ord('a') - shift) % 26 + ord('a'))
            key_index+=1
        else:
            decrypt+=char
    return decrypt

def vernam_encrypt(text, key):
    encrypted = ""
    for i in range(len(text)):
        encrypted += chr( ord(text[i]) ^ ord(key[i]))
    return encrypted

def vernam_decrypt(text, key):
    decrypted = ""
    for i in range(len(text)):
        decrypted += chr( ord(text[i]) ^ ord(key[i]))
    return decrypted


def main():
    msg = "Hello, World!"

    caesar_key = 4
    print("Original Message:", msg)
    encrypted_msg = caesar_encrypt(msg, caesar_key)
    print("Encrypted Message:", encrypted_msg)
    decrypted_msg = caesar_decrypt(encrypted_msg, caesar_key)
    print("Decrypted Message:", decrypted_msg)

    vigenere_key = "key"
    vigenere_encrypted = vigenere_encrypt(msg, vigenere_key)
    print("Vigenere Encrypted Message:", vigenere_encrypted)
    vigenere_decrypted = vigenere_decrypt(vigenere_encrypted, vigenere_key)
    print("Vigenere Decrypted Message:", vigenere_decrypted)

    vernam_key = "qwertwwasdfgo" # Same length as the msg
    vernam_encrypted = vernam_encrypt(msg, vernam_key)
    print("Vernam Encrypted Message:", vernam_encrypted)
    vernam_decrypted = vernam_decrypt(vernam_encrypted, vernam_key)
    print("Vernam Decrypted Message:", vernam_decrypted)
if __name__ == "__main__":
    main()
```

Output:

```
(lab) ak@DevLab:~/Desktop/Data/College/Semester6/Assignment/Crypto Lab/Ex1a$ python3 ./crypto_functions.py
Original Message: Hello, World!
Encrypted Message: Lipps, Asvph!
Decrypted Message: Hello, World!
Vigenere Encrypted Message: Rijvs, Uyvjn!
Vigenere Decrypted Message: Hello, World!
Vernam Encrypted Message: 9     6
N
Vernam Decrypted Message: Hello, World!
(lab) ak@DevLab:~/Desktop/Data/College/Semester6/Assignment/Crypto Lab/Ex1a$
```

Code: (--Caesar Cipher--)

Sender (server.py):

```
import socket
import crypto_functions as cf

key = 4

def encrypt(text):
   return cf.caesar_encrypt(text,key)
def decrypt(text):
   return cf.caesar_decrypt(text,key)

server_socket = socket.socket(socket.AF_INET, socket.SOCK_STREAM)

host = "127.0.0.1"
port = 12345
server_socket.bind((host,port))
server_socket.listen(1)
print(f"Server listening on {host}:{port}")

connection, address = server_socket.accept()
print(f"Connected to {address}")

while True:
   data = connection.recv(1024).decode()
   print(f"Encrypted message from client: {data}")
   data = decrypt(data)
   print(f"Decrypted message from client: {data}\n")

   if data.lower() == "exit":
       print("Exiting...")
       break

   msg = input("You: ")
   enc_msg = encrypt(msg)
   print(f"Encrypted message sent to client: {enc_msg}\n")
   connection.send(enc_msg.encode())

connection.close()
server_socket.close()
```

Receiver (client.py):

```
import socket
import crypto_functions as cf

key = 4

def encrypt(text):
   return cf.caesar_encrypt(text,key)
def decrypt(text):
   return cf.caesar_decrypt(text,key)

client_socket = socket.socket(socket.AF_INET,socket.SOCK_STREAM)

host = "127.0.0.1"
port = 12345
client_socket.connect((host, port))
print("Conected to the server")


while True:
   message = input("You: ")
   message = encrypt(message)
   print(f"Encrypted message sent to server: {message}\n")
   client_socket.send(message.encode())

   if message.lower() == "exit":
       break

   msg = client_socket.recv(1024).decode()
   print(f"Encrypted message from server: {msg}")
   data = decrypt( msg)
   print(f"Decrypted message from server: {data}\n")

client_socket.close()
```

Output:

```
(lab) ak@DevLab:~/Desktop/Data/College/Semester6/Assignment/Cryp
to Lab/Ex1a$ python3 ./server.py
Server listening on 127.0.0.1:12345
Connected to ('127.0.0.1', 59972)
Encrypted message from client: Lipps
Decrypted message from client: Hello

You: Hii, How Are you>?
Encrypted message sent to client: Lmm, Lsa Evi csy>?

Encrypted message from client: Kviex, Lsa efsyx csy>
Decrypted message from client: Great, How about you>

You:
```

```
(lab) ak@DevLab:~/Desktop/Data/College/Semester6/Assignment/Cryp
to Lab/Ex1a$ python3 client.py
Conected to the server
You: Hello
Encrypted message sent to server: Lipps

Encrypted message from server: Lmm, Lsa Evi csy>?
Decrypted message from server: Hii, How Are you>?

You: Great, How about you>
Encrypted message sent to server: Kviex, Lsa efsyx csy>
```

python3
python3

Code: (--Vigenere Cipher--)

Sender (server.py):

```
import socket
import crypto_functions as cf

key = “key”

def encrypt(text):
   return cf.vigenere_encrypt(text,key)
def decrypt(text):
   return cf.vigenere_decrypt(text,key)

server_socket = socket.socket(socket.AF_INET, socket.SOCK_STREAM)

host = "127.0.0.1"
port = 12345
server_socket.bind((host,port))
server_socket.listen(1)
print(f"Server listening on {host}:{port}")

connection, address = server_socket.accept()
print(f"Connected to {address}")

while True:
   data = connection.recv(1024).decode()
   print(f"Encrypted message from client: {data}")
   data = decrypt(data)
   print(f"Decrypted message from client: {data}\n")

   if data.lower() == "exit":
       print("Exiting...")
       break

   msg = input("You: ")
   enc_msg = encrypt(msg)
   print(f"Encrypted message sent to client: {enc_msg}\n")
   connection.send(enc_msg.encode())

connection.close()
server_socket.close()
```

Receiver (client.py):

```
import socket
import crypto_functions as cf

key = “key”

def encrypt(text):
   return cf.vigenere_encrypt(text,key)
def decrypt(text):
   return cf.vigenere_decrypt(text,key)

client_socket = socket.socket(socket.AF_INET,socket.SOCK_STREAM)

host = "127.0.0.1"
port = 12345
client_socket.connect((host, port))
print("Conected to the server")


while True:
   message = input("You: ")
   message = encrypt(message)
   print(f"Encrypted message sent to server: {message}\n")
   client_socket.send(message.encode())

   if message.lower() == "exit":
       break

   msg = client_socket.recv(1024).decode()
   print(f"Encrypted message from server: {msg}")
   data = decrypt( msg)
   print(f"Decrypted message from server: {data}\n")

client_socket.close()
```

Output:

```
(lab) ak@DevLab:~/Desktop/Data/College/Semester6/Assignment/Cryp
to Lab/Ex1a$ python3 ./server.py
Server listening on 127.0.0.1:12345
Connected to ('127.0.0.1', 56126)
Encrypted message from client: Rijvs zbsrrip, Rijvs Zbsrrip, Rij
vs Zbsrrip
Decrypted message from client: Hello brother, Hello Brother, Hel
lo Brother

You: Nice good brother nice nice good good bro
Encrypted message sent to client: Xmao kmyh zbsrrip xmao rgmi ey
sb qsmn fpy

Encrypted message from client: Yoyi, Kmyh Eyh Jyvb Qsmysb
Decrypted message from client: Okay, Good God Lord Gooood

You: []
```

```
(lab) ak@DevLab:~/Desktop/Data/College/Semester6/Assignment/Cryp
to Lab/Ex1a$ python3 client.py
Conected to the server
You: Hello brother, Hello Brother, Hello Brother
Encrypted message sent to server: Rijvs zbsrrip, Rijvs Zbsrrip,
Rijvs Zbsrrip

Encrypted message from server: Xmao kmyh zbsrrip xmao rgmi eysb
qsmn fpy
Decrypted message from server: Nice good brother nice nice good
good bro

You: Okay, Good God Lord Gooood
Encrypted message sent to server: Yoyi, Kmyh Eyh Jyvb Qsmysb
```

```
python3
python3
bash
```

Code: (--Vernam Cipher--)

Sender (server.py):

```
import socket
import crypto_functions as cf

key = “keykey” #the text length must be of length 6

def encrypt(text):
   return cf.verman_encrypt(text,key)
def decrypt(text):
   return cf.verman_decrypt(text,key)

server_socket = socket.socket(socket.AF_INET, socket.SOCK_STREAM)

host = "127.0.0.1"
port = 12345
server_socket.bind((host,port))
server_socket.listen(1)
print(f"Server listening on {host}:{port}")

connection, address = server_socket.accept()
print(f"Connected to {address}")

while True:
   data = connection.recv(1024).decode()
   print(f"Encrypted message from client: {data}")
   data = decrypt(data)
   print(f"Decrypted message from client: {data}\n")

   if data.lower() == "exit":
       print("Exiting...")
       break

   msg = input("You: ")
   enc_msg = encrypt(msg)
   print(f"Encrypted message sent to client: {enc_msg}\n")
   connection.send(enc_msg.encode())

connection.close()
server_socket.close()
```

Receiver (client.py):

```
import socket
import crypto_functions as cf

key = “keykey” # The text buffer must be of length 6

def encrypt(text):
   return cf.verman_encrypt(text,key)
def decrypt(text):
   return cf.verman_decrypt(text,key)

client_socket = socket.socket(socket.AF_INET,socket.SOCK_STREAM)

host = "127.0.0.1"
port = 12345
client_socket.connect((host, port))
print("Conected to the server")


while True:
   message = input("You: ")
   message = encrypt(message)
   print(f"Encrypted message sent to server: {message}\n")
   client_socket.send(message.encode())

   if message.lower() == "exit":
       break

   msg = client_socket.recv(1024).decode()
   print(f"Encrypted message from server: {msg}")
   data = decrypt( msg)
   print(f"Decrypted message from server: {data}\n")

client_socket.close()
```

Output:

```
(lab) ak@DevLab:~/Desktop/Data/College/Semester6/Assignment/Cryp
to Lab/Ex1a$ python3 ./server.py
Server listening on 127.0.0.1:12345
Connected to ('127.0.0.1', 34044)
Encrypted message from client: #)
Decrypted message from client: HeyBud

You: YeaBud
Encrypted message sent to client: 2)

Encrypted message from client:
                                KF
Decrypted message from client: gud u?

You: me gud
Encrypted message sent to client: Y


^[S
```

```
(lab) ak@DevLab:~/Desktop/Data/College/Semester6/Assignment/Cryp
to Lab/Ex1a$ python3 ./client.py
Conected to the server
You: HeyBud
Encrypted message sent to server: #)

Encrypted message from server: 2)
Decrypted message from server: YeaBud

You: gud u?
Encrypted message sent to server:
                                KF

Encrypted message from server: Y

Decrypted message from server: me gud

You:
```

python3
python3
bash